# Sahaj Bhadja

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## Summary

Verification-focused software engineer with 2+ years using **C++** and **Python** across simulation, concurrent processing, cross-platform systems, and automated validation. Experienced in requirements-based testing, debugging, failure isolation, documentation, configuration control, and reliable software design.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Software Engineering, Analysis of Algorithms, Database Systems, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*
- Architected a **Python** application for 500+ users with explicit access controls, indexed data paths, and monitored execution, reducing latency from 1.2 seconds to 350 milliseconds through systematic performance analysis.
- Verified fault-tolerant asynchronous workers processing 1,000+ jobs through explicit state transitions, failure injection, structured logging, and recovery-path testing, increasing throughput threefold while preventing cascading failures under production load.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*
- Built Python services handling 10,000+ daily requests, implementing secure ingestion, validation, error handling, and repeatable functional tests while maintaining stable behavior and observable performance during peak production load.
- Developed a **multithreaded** pipeline for 1,000+ files using ThreadPoolExecutor, testable state handling, fault recovery, progress monitoring, and repeatable defect reproduction, reducing end-to-end processing turnaround by 60%.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*
- Built a Python **simulation and verification pipeline** processing 100,000+ trajectory records, improving planning quality 18% through reproducible test scenarios, automated validation, performance tracing, and controlled software configuration.
- Executed closed-loop functional verification across complex sensor scenarios, improving simulation stability 15% while documenting expected behavior, isolating failures, comparing outputs, and debugging interactions between software components systematically.

**Aasma Technology Solutions** **India**
*Software Developer Engineer – SecureSync* *August 2023 – July 2024*
- Engineered cross-platform software with automated executable builds and VM-based verification across Windows, Linux, and macOS, maintaining 20+ unit and integration test suites, coverage reports, and failure logs.
- Reviewed and improved object-oriented integration modules through requirements analysis, code changes, regression testing, and defect tracking, increasing processing efficiency 68% while reducing technical debt 20% through maintainable design standards.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern – VibrantChem* *May 2023 – August 2023*
- Developed TypeScript and Node.js software across eight microservices, implementing defined interfaces and data-conversion workflows to synchronize more than two million records across real-time operations and modular service boundaries.
- Automated functional, integration, and regression testing with Jenkins, Docker, Jest, Mocha, Selenium, Sentry, and Azure monitoring, cutting deployment time 50% while improving release verification and change traceability.

## Skills

**Languages & Systems:** C++, Python, Java, Bash, Linux, object-oriented programming, data structures, algorithms, operating systems, concurrency, multithreading
**Verification:** Functional verification, test coverage reporting, requirements-based testing, unit testing, integration testing, regression testing, performance testing, traceability, change tracking
**Testing & Debugging:** PyTest, Mockito, Jest, Selenium, test planning, automated test programs, defect reproduction, fault isolation, error handling, logging, monitoring
**Development & Tools:** Git, Docker, Jenkins, Maven, CI/CD, build automation, VM-based testing, simulation, cross-platform development, coding standards, technical documentation, code reviews

## Projects

**CPU-Scheduling Simulator** | Live | *C++, React, Operating Systems, EJ-2*
- Implemented six deterministic CPU-scheduling algorithms in **C++** with React controls and Gantt visualization, validating 1,000+ process combinations while reducing average turnaround time 25% and debugging time 20%.

**Chat2SQL** | Code | *Java, TypeScript, Docker, Maven, SQL*
- Built a cross-platform Java and TypeScript application supporting 10+ databases, schema-aware query generation, structured validation, automated testing, and Docker-based deployment across Windows, macOS, and Linux development environments.

**FinNexus** | Live | *Real-Time Systems, WebSockets, Kubernetes, MongoDB*
- Built a real-time platform supporting 50,000+ users and one million transactions, reliably integrating WebSocket updates, failure monitoring, Kubernetes deployment, automated releases, and performance-tuned services under high-volume workloads.